HRITHIK GHOSH

AI & ML Engineer

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Summary

Software Engineer with 4+ years of experience in backend systems, scalable product engineering, and AI-driven solution development. My background spans Java, Spring Boot, databases, and system design, with prior industry experience at Amdocs delivering performance optimization, API development, testing automation, and reliability improvements in production environments.

More recently, through my work with ITU, I have expanded into applied AI and healthcare-focused intelligent systems, contributing to the design and development of multilingual, clinically aware AI platforms. My experience includes architecting Retrieval-Augmented Generation pipelines, semantic caching strategies, multimodal voice-and-text systems, frontend interfaces, and evaluation-oriented AI workflows for real-world public health use cases. I have also worked on system architecture documentation, pipeline design, and team onboarding processes to support scalable and independent project delivery.

Currently pursuing a Master's degree in AI & Data Science, I am deeply committed to building practical, trustworthy, and impactful AI systems. With a strong software engineering foundation and growing expertise in machine learning, NLP, LLM systems, and applied AI architecture, I aim to contribute to scalable, responsible, and high-impact AI solutions that solve real-world problems.

Experience

International Telecommunication Union — Geneva, Switzerland

Backend Dev & AI Full Stack engineer (LLM & Voice Module) — 03/2026

- Conducted an end-to-end architectural review of the Genie AI NCD Healthcare pipeline and re-engineered the semantic cache intercept point post-guardrail, eliminating redundant LLM inference on cache hits and significantly reducing API token costs and response latency
- Created a clinical query classification gate to route safety-critical NCD queries via the full RAG pipeline. Directed general queries using cosine similarity (⩾0.95 clinical, ⩾0.85 general), avoiding incorrect cached responses for distinct but similar queries.
- Developed a retrieval pipeline integrating vector database search, sparse indexing, Knowledge Graph, and keyword search to deliver accurate, guideline-based answers on cache misses.
- Developed a multi-path retrieval pipeline integrating vector search, sparse indexing, Knowledge Graph traversal, keyword search, filtering, and Re-Ranker to deliver contextually accurate, guideline-based answers on cache misses.
- Architectured a validated cache store-back mechanism that persists only guardrail-approved LLM responses paired with query embeddings, enabling progressive cache warming while guaranteeing clinically safe responses are served on future cache hits
- Defined a TTL-based cache invalidation strategy linked to knowledge base versioning, automatically purging cached clinical responses when source documents or medical guidelines are updated
- Integrated a multilingual voice-text I/O system. Connected STT transcription, TTS synthesis, and Machine Translation layers. Ensured consistent cache behavior across input languages.
- Designed and developed the Amina Care frontend interface, implementing voice and text input/output components, session-aware chat UI, and an interactive pipeline visualisation layer to support both end users and internal team workflows
- Produced system architecture documentation, redesigned pipeline diagrams, and authored full team onboarding materials covering server SSH access, GitLab PAT setup, branch configuration, and daily Git workflow

Amdocs — Pune, India

Software Engineer — 06/2021 - 10/2023

- Developed and maintained Amdocs products (CRM, OMS) with Java and Spring, achieving a 20% boost in application performance
- Executed thorough testing for online/offline events and billing, ensuring a 95% accuracy rate in functionality
- Conducted API testing and debugging for Java and REST APIs, decreasing bug resolution time by 30%
- Streamlined development processes by leveraging technologies like Ginger, JSON, and CI/CD pipelines, enhancing team efficiency
- Helped with frontend development with Rest API and React JS
- Automated testing processes with Ginger and Selenium, resulting in a 40% reduction in manual testing time

West Bengal Youth Computer Center (Jagacha) — Kolkata, India

Software Engineer — 06/2025 - 06/2025

- Worked on Front-end Web Development with the HP exam integration system to the youth centers exam system
- Designed Workflow and CI-CD integrations with Jira
- Worked with SQL to merge student dataset with billing entities
- Worked with students and professionals with mentorship programs

Education

KIIT Institute of Technology (KIIT) — Bhubaneshwar, India

Masters in Technology — 2025 - Ongoing

- Specialization - AI & DS - SGPA 8.34

Education

Maulana Abul Kalam Azad University of Technology, West Bengal formerly WBUT — Kolkāta, India

Bachelors in Technology — 2017 - 2021

- Specialization - Computer Science & Engineering - CGPA 7.92

Projects

Agentic-rag-evaluator-v2

https://github.com/Hrithik373/ARAassistant — 01/2026

A production-ready Agentic Retrieval-Augmented Generation (RAG) system with document-grounded evaluation, built using LangChain, FAISS, OpenAI LLMs, and Streamlit.

- Built a production-ready Agentic RAG evaluation system using LangChain, FAISS, OpenAI LLMs, and Streamlit to deliver document-grounded question answering over uploaded PDF knowledge bases.
- Developed an end-to-end retrieval pipeline that ingests PDFs, performs chunking and vector indexing, retrieves relevant context, and enables agentic LLM reasoning with conversation-aware execution.
- Implemented an LLM-as-Judge evaluation framework to score generated responses on relevance, faithfulness, groundedness, latency, and overall answer quality.
- Designed and deployed a cloud-ready Streamlit interface with a modern metrics dashboard, responsive UX, and interactive evaluation visualizations, optimized for lightweight CPU-based deployment.

ML-Driven SMS Spam Detection Engine — Kolkāta, India

https://github.com/Hrithik373/sms-spam-detection — 2025

- The project uses **Python**, with **Pandas/NumPy** for preprocessing and **NLTK/spaCy** for text cleaning, tokenization, stop-word removal, and TF-IDF vectorization. Machine-learning models such as **Logistic Regression, Naïve Bayes, SVM, Random Forest**, and **XGBoost** were trained using **Scikit-learn**. Visualisations were created through **Matplotlib/Seaborn**, and the final model was optionally deployed using **Flask/Streamlit** for interactive prediction.
- Enhanced performance with data cleaning, optimized TF-IDF, and n-gram features. Managed imbalance with SMOTE, tuned class-weight. Used GridSearchCV, cross-validation, and tested models like SVM, XGBoost. Added logging, modular code, and metrics (F1, ROC-AUC) for robustness.

Retail Sales Forecasting System

https://github.com/Hrithik373/Retail-Prediction-model — 2025

Built a retail sales forecasting system using Python, Pandas/NumPy for data prep. Explored models: Linear Regression, Random Forest, XGBoost, ARIMA, LSTM. Visualized with Matplotlib/Seaborn. Developed forecasting pipeline via Scikit-learn/TensorFlow/Keras.

- Enhanced accuracy via refined feature engineering: lag features, rolling stats, seasonality, event signals. Applied hyperparameter tuning and cross-validation. Leveraged XGBoost, LSTM for non-linear patterns. Integrated into production pipeline with MAPE, RMSE, R^2 metrics for retail forecasts.

Spotify-clone — Kolkāta, India

https://github.com/Hrithik373/Spotify-a — 2022

- Developed a React.js, Node.js web app, enhancing user engagement
- Boosted data handling efficiency by 30% using Python APIs, MongoDB
- Integrated Spotify API, increasing user retention by 25%

Face Recognition and Emotion Detection Model — Kolkāta, India

https://github.com/Hrithik373/Emotion-Detection-Model — 2021

Final Year Project: "Face Recognition and Emotion Detection" using CNN model, 85% accuracy . Engineered recognition feature with Raspberry Pi, OpenCV. Utilized Python, Django to cut processing time by 30%.

- What was a successful outcome of your work? (e.g. Raised $3,000 for the charity)

Technical Stack

Languages: Python · Java (beginner) · ReactJS · VueJs · Three JS · Typescript · CSS · JavaScript (Beginner)

APIs & Tools: Postman · FastAPI · RestApi · GraphQL

ML&DS Skills

Python Libraries: - NumPy · Pandas · Matplotlib · Seaborn

Frameworks: Scikit-learn · XGBoost · Lightgbm

Deep Learning: TensorFlow · Transformers (BERTs) · CNN

ML&DS Skills

LLM, Microservices: Whisper STT · ConquiTTS · Piper TTS · Mistrol · OpenAI · Google Flan

Vector Database: Arcade DB · ArangoDB (Only existing intgration)

Integration Tools: HayHooks · Kong

Architectural Skills: RAG · OPEA (only leaned the existing OPEA part) · Haystack

Tools: Docker · Kubernetes

Computer Vision: OpenCV (beginner)

DATABASES, DEPLOYMENT TOOLS

Deployment Tools : VS Code, Jupyter, Google Colab, Streamlit

Data Bases : SQL, MongoDB

Certifications

Udemy — The Complete Python Course | Learn Python by Doing in 2025

Kaggle Deep Learning Certification — Kaggle Certified

Kaggle Machine Learning Certification — Kaggle Certified

Languages

English Proficient ●●●●○

Hindi Advanced ●●●○○

Bengali Native ●●●●●

Find me online

Github
https://github.com/Hrithik373